

1.1 Linear and Rational Equations

Definition

A **linear equation in one variable** is an equation that can be expressed in the form

$$ax + b = 0,$$

where a and b are real numbers with $a \neq 0$, and x is the variable.

Linear equations are among the simplest algebraic relationships. Solving such an equation means finding the value of x that makes both sides equal, balancing constants and coefficients. The structure $ax + b = 0$ represents a straight line when graphed, and the point where it crosses the x -axis corresponds to its solution.

Example 1. Solve $3x - 9 = 0$.

$$3x = 9 \Rightarrow x = 3.$$

Example 2. Solve $5x + 7 = 0$.

$$5x = -7 \Rightarrow x = -\frac{7}{5}.$$

Example 3. Solve $-4x + 12 = 0$.

$$-4x = -12 \Rightarrow x = 3.$$

A practical use of linear equations often appears in finance or planning situations. For instance, consider comparing the total cost of two options over time.

Example. A couple must decide whether to buy a used car for \$8800 or lease a new one for an initial down payment of \$2500 followed by monthly payments of \$225.

(a) Write a cost model C (in dollars) to lease the car for t months. (b) Determine when the cost to lease equals the cost to buy.

Solution. The total cost combines the initial payment and the recurring monthly payments:

$$C = 225t + 2500.$$

To find when leasing equals buying:

$$8800 = 225t + 2500,$$

$$8800 - 2500 = 225t,$$

$$6300 = 225t,$$

$$t = 28.$$

Thus, after 28 months, both options cost the same.

Definition

A **conditional equation** is true for some, but not all, values of the variable. An **identity** is true for every possible value of the variable. A **contradiction** is false for every value of the variable.

These categories describe how broadly an equation holds. A conditional equation restricts possible solutions; an identity expresses a universal truth (like algebraic laws), and a contradiction signals an impossibility.

Example 1. $2x + 3 = 7$ is conditional since $x = 2$ is its only solution.

Example 2. $3(x + 2) = 3x + 6$ is an identity—it's true for all x .

Example 3. $x + 4 = x + 2$ is a contradiction—no x satisfies it.

Definition

A **rational equation** is an equation in which each term is a rational expression—an expression that can be written as a fraction of polynomials.

All linear equations can be viewed as rational equations because they may be written as fractions (for example, dividing both sides by a coefficient). However, not every rational equation is linear. Rational equations sometimes include variables in the denominator, and in those cases, we must be careful to avoid division by zero.

Linear Equation with Constants in Denominator:

$$\frac{x}{2} = \frac{2x}{3} - 1$$

Rational Equation with Variables in Denominator:

$$\frac{12}{x} = \frac{6}{2x} + 3$$

The same technique used for solving linear equations applies: multiply through by the least common denominator (LCD) to remove fractions. When variables appear in denominators, we must note values that make the denominators zero, since those are excluded from the solution set.

Example 1. Solve $\frac{1}{x} = \frac{2}{3}$. Multiply both sides by $3x$: $3 = 2x \Rightarrow x = \frac{3}{2}$.

Example 2. Solve $\frac{x}{x-1} = 2$. Multiply by $x-1$: $x = 2x-2 \Rightarrow x = 2$.

Example 3. Solve $\frac{3}{x+2} + 1 = \frac{6}{x+2}$. Multiply by $x+2$: $3 + x + 2 = 6 \Rightarrow x = 1$.

Example. Solve and check:

$$\frac{x}{x-4} = \frac{4}{x-4} - \frac{4}{5}.$$

Solution. First, eliminate denominators by multiplying both sides by $5(x-4)$:

$$5x = 20 - 4(x-4).$$

Simplify:

$$5x = 20 - 4x + 16 \quad \Rightarrow \quad 9x = 36 \quad \Rightarrow \quad x = 4.$$

However, note that $x = 4$ makes the denominator $x-4 = 0$, which is undefined. **Therefore, there is no valid solution.**

Example 1. $\frac{x}{x+3} = \frac{6}{x+3} - 2$. Multiply by $x+3$: $x = 6 - 2x - 6 \Rightarrow 3x = 0 \Rightarrow x = 0$.

Example 2. $\frac{3x}{x-2} = \frac{9}{x-2} - 3$. Multiply by $x-2$: $3x = 9 - 3x + 6 \Rightarrow 6x = 15 \Rightarrow x = \frac{5}{2}$.

Example 3. $\frac{x}{x+1} = 1 - \frac{2}{x+1}$. Multiply by $x+1$: $x = x+1-2 \Rightarrow 1 = 2$ — a contradiction. **No solution.**

Real-World Example: Commuting Costs

A commuter rides the subway to work. Each ride costs \$2.50.

- (a) Write a model for the total cost C (in dollars) for x rides.
- (b) If the commuter can buy an unlimited-ride MetroCard for \$105 per month, how many rides make it worthwhile to buy the pass?

Solution:

- (a) The cost is directly proportional to the number of rides: $C = 2.50x$.
- (b) To find when the pass saves money, set $C = 105$:

$$105 = 2.50x \quad \Rightarrow \quad x = 42$$

So, after 42 rides, the MetroCard becomes the better deal.